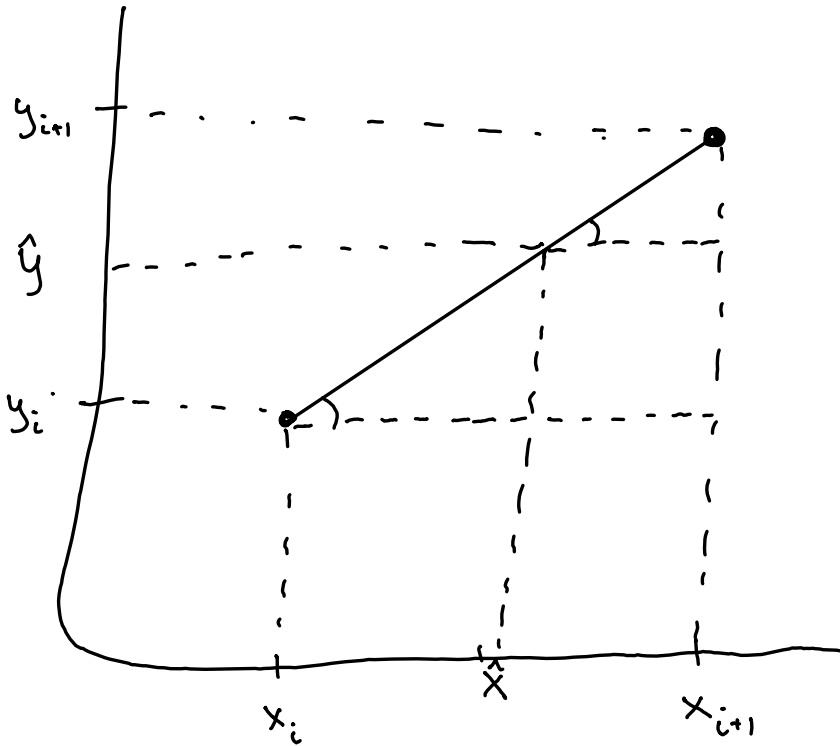


02-24-2025

$$\frac{\hat{y} - y_i}{\hat{x} - x_i} = \frac{y_{i+1} - y_i}{x_{i+1} - x_i} \Rightarrow \frac{\hat{y} - y_i}{y_{i+1} - y_i} = \frac{\hat{x} - x_i}{x_{i+1} - x_i} = \omega$$

$$\Rightarrow \hat{y} = (1 - \omega) y_i + \omega \cdot y_{i+1}$$



$$\max_{k'} \{ u(zf(k) - k') + \beta v^{(0)}(k') \}$$

$$k' \in [k_j, k_{j+1}] ; v^{(0)}(k') = (1 - \omega) v^{(0)}(k_j) + \omega v^{(0)}(k_{j+1})$$

$$\omega = \frac{k' - k_j}{k_{j+1} - k_j}$$

Backward Induction:

Working:

$$V(k_t, x_t, z_t, t) = \max_{c_t, k_{t+1}, l_t} \{ u(c_t, l_t) + \beta \sum_{x_{t+1}} \sum_{z_{t+1}} \Psi(x_t | x_{t+1}) T(z_t | z_{t+1}) V' \}$$

s.t.

$$k_{t+1}(1+r) + wz_t l_t = k_{t+1} + c_t + x_t + Tr_t$$

$$Tr_t = \min \{ 0, c_{\min} - k_t(1+r) - wz_t l_t + x_t \}$$

$$\Rightarrow k_{t+1} = 0$$

Retired:

$$VR(k_t, x_t, t) = \max_{c_t, k_{t+1}} \{ u(c_t, 0) + \beta \sum_x \Psi(x_t | x_{t+1}) VR'(x', c', t) \}$$

s.t.

$$k_t(1+r) + ss = k_{t+1} + c_t + x_t + Tr_t$$

$$Tr_t = \min \{ 0, c_{\min} - k_t(1+r) - ss + x_t \}$$

Solve:

① Discretize State Space: k, x, z ; $kgrid(1:n)$
 $xgrid(1:m)$
 $zgrid(1:v)$

② Initialize Value & Policy Functions:

$VR(1:n, 1:m, R:T)$, $V(1:n, 1:v, 1:m, 1:R-1)$

$coptR(1:n, 1:m, R:T)$; $copt(1:n, 1:v, 1:m, 1:R-1)$

$koptR(1:n, 1:m, R:T)$; $kopt(1:n, 1:v, 1:m, 1:R-1)$
 $lopt(1:n, 1:v, 1:m, 1:R-1)$

Start with age T:

for $ik = 1:n$

for $ix = 1:m$

$koptR(ik, ix, T) = 0$;

$totres = kgrid(ik)(1+r) + ss - xgrid(ix)$;

if $totres < cmin$

$coptR(ik, ix, T) = cmin$;

$VR(ik, ix, T) = \log(cmin)$;

else

$coptR(ik, ix, T) = totres$;

$VR(ik, ix, T) = \log(totres)$;

end % end if clause

end % end x loop

end % end k loop

For $t = T-1 : R$

for $ik = 1 : n$

for $ix = 1 : m$

$totres = kgrid(ik)(1+r) + ss - xgrid(ix)$

if $totres < cmin$

$coptR(ik, ix, t) = cmin$

for $ixp = 1 : n$

$val = val + \psi(ixp, ix) VR(1, ixp, t+1)$

end

$VR(ik, ix, t) = \log(cmin) + \beta val$

$koptR(ik, ix, t) = 0$

else if $totres > min$

↳ new file ufunc (next page)

$\sum_{i=1}^3 p_i x_i$: $val = 0$
for $i = 1 : 5$
 $val = val + p_i x_i$
end

function $U = ufunc(kp, ik, ix,$
global $kgrid, r, ss, xgrid, \beta, \psi(,), VR$

$$cons = kgrid(ik)(1+r) + ss - xgrid(ix) - kp;$$

$$Util_{today} = \log(cons);$$

$$wght = \frac{kp - kgrid(jl_0)}{kgrid(jl_{on}) - kgrid(jl_0)};$$

$$Eval = 0;$$

for $ixp = 1:m$

$$V_{int} = (1 - \omega)VR(jl_0, ixp, t+1) + \omega VR(jl_{t+1}, ixp, t+1);$$

$$Eval = Eval + \psi(ixp, ix) V_{int};$$

end

$$U = Util_{today} + \beta Eval;$$

$$U = -U$$